CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rectangular field 36 m by 14 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 252 **(b)** 266 **(c)** 216 **(d)** 288

2. A picture 35 cm by 24 cm is mounted on a frame of width 2 cm all around. The area of the frame alone (cm^2) is: 1

(a) 254 **(b)** 250 **(c)** 260 **(d)** 252

3. A wire in the shape of a square of side 30 cm is straightened and bent into a rectangle of length 39 cm. The breadth (cm) is: 1

(a) 39 **(b)** 9 **(c)** 21 **(d)** 12

4. Tiling a rectangular floor of length 15 m at ₹120 per m^2 costs ₹10800. Its breadth (m) is: 1

(a) 6 **(b)** 12 **(c)** 21 **(d)** 90

5. Out of all rectangles with perimeter 48 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 144 **(b)** 168 **(c)** 132 **(d)** 156

6. The sides of a triangle are 21 cm, 28 cm and 35 cm. Its perimeter (cm) is: 1

(a) 84 **(b)** 105 **(c)** 85 **(d)** 83

7. A rectangle's length is 3 times its breadth and its perimeter is 88 cm. Its area (cm^2) is: 1

(a) 385 **(b)** 352 **(c)** 363 **(d)** 374

8. A square and an equilateral triangle have equal perimeters. If the square's side is 27 cm, the triangle's side (cm) is: 1

(a) 63 **(b)** 27 **(c)** 45 **(d)** 36

9. **Assertion (A):** The area of a triangle with base 10 cm and height 6 cm is 30 cm^2. 1
Reason (R): Area of a triangle = $\frac{1}{2}$ x base x height.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** To fence a field we find its perimeter, not its area. 1
Reason (R): A fence goes along the boundary of the field.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A carpet 7 m by 4 m is laid in a room 10 m by 6 m. Find the area of the uncovered floor. 2

12. The length and breadth of a rectangle are in the ratio 7:5 and its perimeter is 72 cm. Find the sides. 2

13. A rectangular field is 35 m by 25 m. Find the cost of levelling the field at ₹6 per m^2 and fencing at ₹8 per metre. 2

14. A carpet 8 m by 6 m is laid in a room 12 m by 9 m. Find the area of the uncovered floor. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular plot 70 m by 40 m is to be fenced with 2 rounds of barbed wire costing ₹20 per metre. Find the total cost. Also find the cost of ploughing it at ₹2 per m^2. 3

16. A square of side 25 cm and a rectangle of length 31 cm and breadth 19 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

17. A floor 12 m by 5 m is to be covered with tiles each 40 cm by 30 cm. Find the number of tiles. 3

OR

A rectangular plot 55 m by 30 m is to be fenced with 3 rounds of barbed wire costing ₹14 per metre. Find the total cost. Also find the cost of ploughing it at ₹4 per m^2. 3

18. A square of side 16 cm and a rectangle of length 23 cm and breadth 9 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

OR

A floor 7 m by 6 m is to be covered with tiles each 35 cm by 20 cm. Find the number of tiles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A garden bed is in the shape of a right-angled triangle. Its two shorter sides are 12 m and 9 m and its longest side is 15 m. 5

(i) Find its perimeter.

(ii) Find its area.

(iii) What is the area of a rectangle 12 m by 9 m?

(iv) Edging costs ₹30 per metre. Find the cost of edging the bed all round, and say how the area of the bed compares with the rectangle.

20. A rectangle with length 28 cm and breadth 15 cm has a triangular piece with base 28 cm and height 10 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A rectangular park 50 m by 30 m has two paths each 2 m wide crossing in the middle, one parallel to each side. Find the area of the paths and the area of the rest of the park. 5

END OF QUESTION PAPER — **Total: 40 Marks**